

Wearables and Habits: Staggered DiD, Placebos and Shoe Bundling

Issue #7

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1 Summary

Exploratory staggered DiD (Callaway–Sant’Anna, never-treated controls, no covariates, user-quarter panel, outcome = any purchase in the quarter) of first wearable purchase on running-related and other purchases. All event studies use the universal base period, so the **reference period** is $t = -1$ (not $t = 0$); the coefficient at -1 is zero by construction.

- **Pre-trends are strong, so the reduced form is not credible as is.** Treated users purchase *less* than controls 2–8 quarters before adoption, relative to $t = -1$, in almost every outcome (Figure 1), including the placebos (Figure 2). Treated users’ purchasing is ramping up into adoption (consistent with the rising pre-adoption purchasing noted in #5).
- **No detectable post-adoption increase in running purchases.** Post ATT (quarters 1–8) for running shoes is -0.016 (0.009), running-related -0.033 (0.016), fitness gear -0.020 (0.014), supplements $+0.022$ (0.017), medication -0.005 (0.014). Negative post estimates relative to $t = -1$ mostly reflect the $t = -1$ peak in a ramp-up, not a causal drop.
- **Placebos mostly small, but the general-intensity outcomes are not.** Pet, books, gift cards, cables, cleaning, grocery and (conditional) shoe price are all within about 2 SE of zero post, yet show the same pre-trend as the main outcomes. Total purchases $\ln(1+n)$ and spend fall post (-0.11 , -0.20) after a pronounced run-up ($t = 0$ spike of $+0.20$ in $\ln(1+n)$ from the device order itself).
- **Bundling is thin.** Only 30 of the 570 adopters buy a running shoe within ± 1 month of the device (24 of 447 clean). Excluding the bundled shoe, no outcome differs significantly between bundlers and non-bundlers except fitness gear (difference -0.144 , SE 0.072, borderline), and the bundler estimates are very imprecise.
- **Running shoes as the treatment:** buying running shoes does not predict a later wearable purchase ($+0.001$, SE 0.007), but does predict later non-shoe running-related purchases ($+0.029$, SE 0.014; $+0.041$ among users who never buy a wearable), with the caveat of pre-trends. Repeat shoe purchase is $+0.082$ (mechanical replacement).

2 Design

- **Panel:** user-quarter over each user’s observed window, truncated at 2023-03 (usable window from #5); zeros filled; unbalanced panel. Built by `code/scripts/prep_design_panel.py`; estimated by `code/scripts/design_did.R`; figures/tables by `design_did_figs.py`.
- **Treated:** first non-kids wearable (Category WEARABLE_COMPUTER/BIOMETRIC_MONITOR) with ≥ 12 months pre and post (570 users); “clean” variant (447 users) also requires price $\geq \$40$,

quantity 1, not an accessory; non-clean adopters are dropped from the clean sample. **Controls:** users who never buy a wearable (about 4,100). Adopters with <12m pre/post are dropped.

- **Estimator:** `did::att_gt`, `est_method="reg"`, no covariates, universal base period, analytic SEs clustered by user, event window -8 to $+8$ quarters. “Post ATT” is the average of $e = 1, \dots, 8$ (excludes $e = 0$, which contains the device purchase itself).
- **Outcomes:** indicators for any purchase in the quarter: running shoe, running-related keyword, fitness gear, supplements, medication (definitions from #5). **Placebos:** pet, books, gift cards, cables/chargers/batteries, cleaning agents, grocery/food; $\ln$ average price of shoes (SHOES category, only quarters with a shoe purchase); and general intensity $\ln(1 + n)$ and $\ln(1 + \text{spend})$.
- **Bundling:** `bundler` = a running shoe bought within ± 1 month of the first wearable. For adopters the bundled shoe is removed from the shoe and running-related outcomes.
- **Shoe as treatment:** first running shoe (604 users with ≥ 12 m pre and post) vs. never-running-shoe users, outcomes wearable, non-shoe running-related, fitness gear, supplements, repeat shoe, with a version dropping all wearable adopters.

3 Main outcomes and placebos

Outcome	Post ATT (SE)	Pre-test p	N treated
Running shoe	-0.016 (0.009)	n/a	570
Running-related (keyword)	-0.033 (0.016)	0.000	570
Fitness gear	-0.020 (0.014)	0.000	570
Supplements	0.022 (0.017)	0.000	570
Medication	-0.005 (0.014)	0.000	570
PLACEBO: pet	-0.024 (0.016)	0.000	570
PLACEBO: books	-0.001 (0.019)	0.000	570
PLACEBO: gift cards	0.002 (0.014)	0.000	570
PLACEBO: cables/batteries	0.004 (0.019)	0.000	570
PLACEBO: cleaning	0.021 (0.014)	0.000	570
PLACEBO: grocery/food	0.020 (0.015)	0.000	570
PLACEBO: $\ln$ shoe price (cond.)	-0.046 (0.068)	0.000	441
General: $\ln(1 + \# \text{purchases})$	-0.110 (0.034)	0.000	570
General: $\ln(1 + \text{spend})$	-0.203 (0.053)	0.000	570

The pre-test column is the joint Wald test of the pre-period ATTs from `did`; it is 0.000 almost everywhere and n/a when the covariance matrix is singular, so rely on the plots. The clean-sample estimates are similar (Table in `design_tab_main_clean.tex`; running-related -0.027 (0.018), shoe -0.011 (0.011)).

4 Running-shoe bundling

Outcome	Bundlers (N=30)	Non-bundlers (N=540)	Difference
Fitness gear	-0.156 (0.071)	-0.012 (0.014)	-0.144 (0.072)
Running-related (ex-bundle)	-0.113 (0.083)	-0.016 (0.016)	-0.098 (0.084)
Running shoe (ex-bundle)	-0.005 (0.053)	-0.014 (0.009)	0.008 (0.054)
Supplements	-0.028 (0.073)	0.024 (0.017)	-0.052 (0.075)

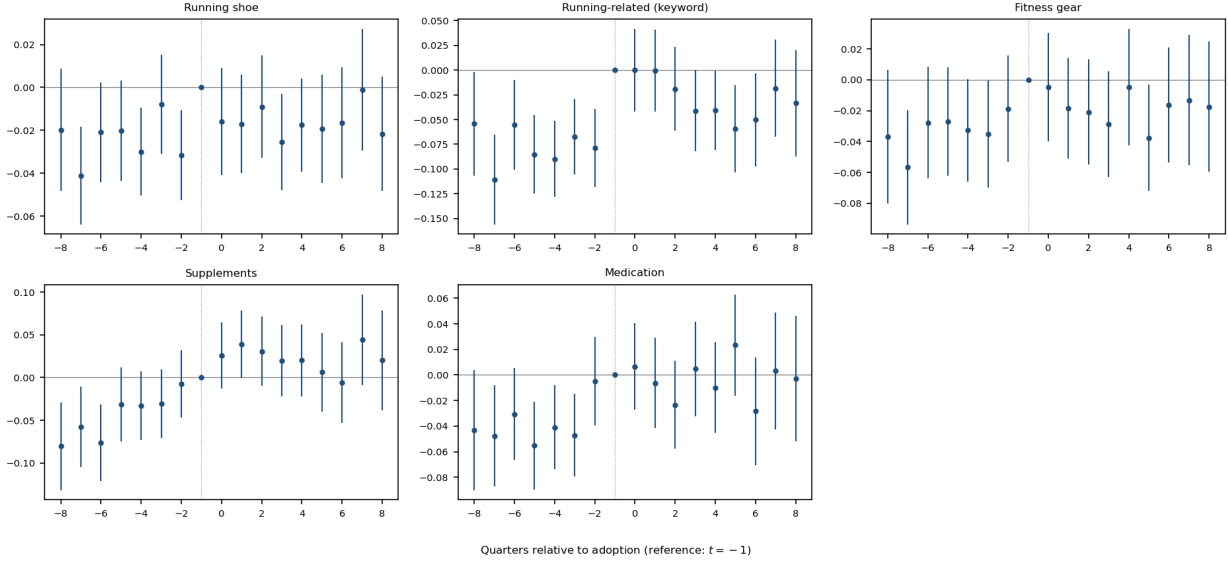


Figure 1: Event studies, main outcomes (570 adopters). 95% CIs; reference $t = -1$.

Post ATT (SE); the difference SE assumes independence (shared controls make it slightly conservative or not, unchecked). Pre-test for bundlers is not available (30 users).

5 Running shoes as the treatment

Outcome	Post ATT (SE)	Pre-test p	N treated
Wearable purchase	0.001 (0.007)	n/a	604
Running-related, non-shoe	0.029 (0.014)	n/a	604
Fitness gear	0.001 (0.013)	0.000	604
Supplements	-0.003 (0.017)	0.000	604
Running shoe (repeat)	0.082 (0.005)	n/a	604
PLACEBO-ish: medication	-0.002 (0.013)	n/a	604
PLACEBO: pet	-0.019 (0.017)	0.000	604
PLACEBO: books	-0.035 (0.018)	0.000	604

Dropping wearable adopters (399 treated): non-shoe running-related +0.041 (0.015), repeat shoe +0.074 (0.006); the wearable outcome is undefined there by construction.

6 Hypotheses for the common pre-trend (untested)

The pre-trend appears in running outcomes, supplements, medication and all placebos alike, and total purchases rise by roughly 0.35 log points from $t = -8$ to $t = -1$. That pattern points to a latent factor driving overall purchasing intensity rather than anything specific to running. Candidate explanations, with what each predicts and how it could be checked with the current data:

1. **Conditioning on activity (selection on the observation window).** Windows run from a user's first to last purchase, and adopters must have ≥ 12 months after the device. Treated users therefore are active at $t = 0$ and stay active, while the never-treated pool includes

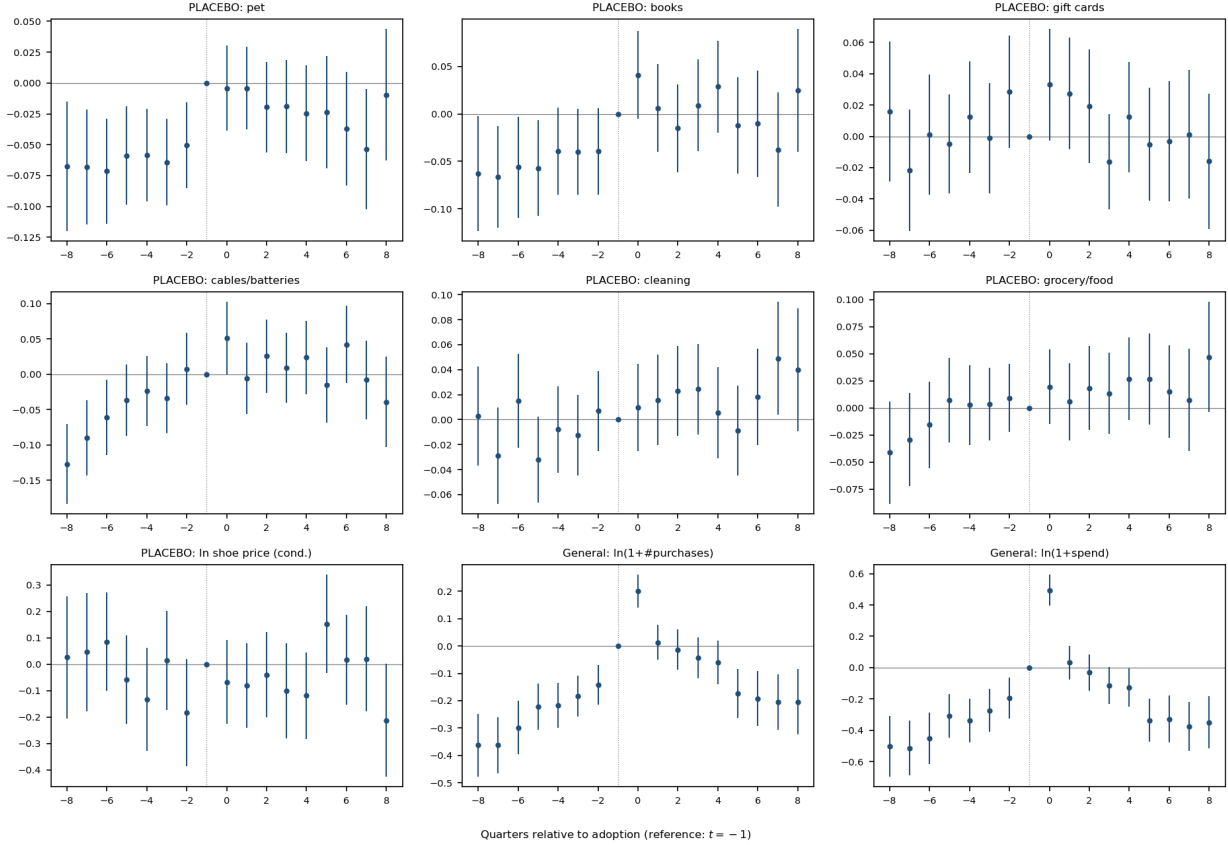


Figure 2: Event studies, placebo and general-intensity outcomes. Placebos show the same pre-trend as the main outcomes.

- dormant or short-window users. This mechanically produces a ramp-up into adoption and no matching ramp in controls. *Check:* restrict controls to users with comparable window length and activity, or match on pre-period purchase counts.
2. **Account/data maturity.** Early quarters after a user's first purchase may reflect a newly active or newly shared account, with purchasing growing over time. *Check:* plot outcomes in time since the user's first purchase for adopters and controls.
 3. **Latent health or fitness intent (anticipation).** A user who starts a fitness or health routine raises supplements, gear, medication and later the device, so purchases rise before the device. This is the plan's main threat; it would show up in health-related outcomes but should be weaker in placebos, which is not what we see, so it is unlikely to be the whole story.
 4. **Life-cycle or life-event shocks.** Moves, a new job, a new baby, a household change or New Year resolutions raise purchases across all categories and also raise the odds of buying a device. *Check:* the survey has household size and demographics; look at event studies by household type, and at adoption by quarter-of-year.
 5. **Income and tech affinity.** Adopters have higher income and larger households (#5), so they may be on a steeper growth path in Amazon spending. *Check:* covariate-adjusted or doubly-robust estimation, or income-specific trends.
 6. **Gifting and seasonality.** Devices are often bought or gifted around holidays, and holiday-heavy shoppers differ in seasonal intensity. Calendar time is handled by the control group,

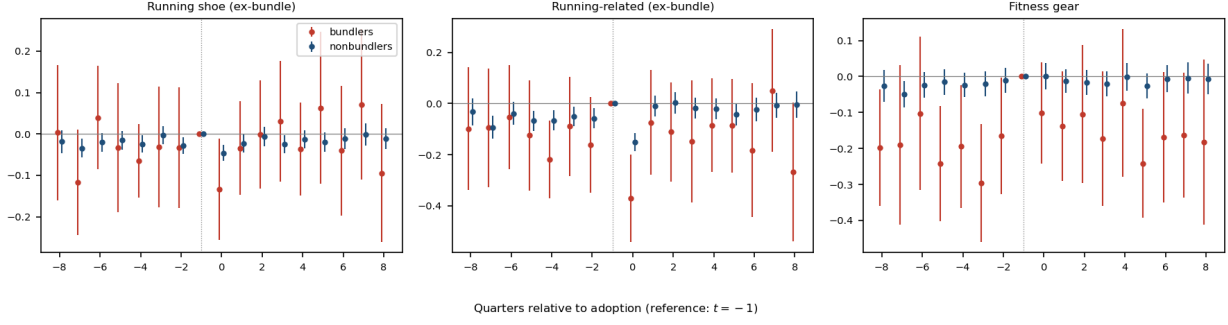


Figure 3: Bundlers (red, 30) vs non-bundlers (blue, 540), bundled shoe excluded from outcomes.

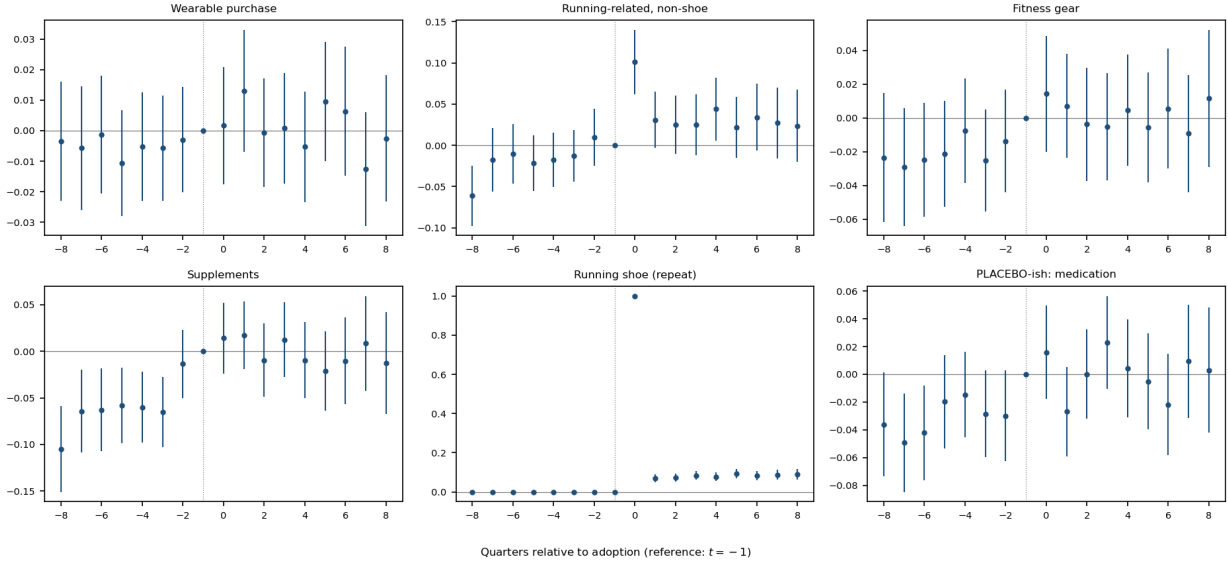


Figure 4: Event studies with the first running-shoe purchase as treatment.

but heterogeneity in seasonal intensity is not. *Check:* adoption timing by quarter and drop Q4 adopters.

7. **Measurement.** Missing Category/Title falls from 10% in 2018 to 1% in 2022, so keyword outcomes are undercounted early; this matters if adopters are observed later in calendar time than controls. The $t = 0$ spike in $\ln(1 + n)$ also reflects the device order itself.

Hypotheses 1 and 2 are mechanical and would need to be ruled out before reading any of the post-adoption estimates as behavioral. Because placebo categories move with the main outcomes, a “purchasing intensity” control (or outcomes expressed as shares of purchases) is the most direct next step.

7 Balanced-panel robustness

Balanced panel: only users observed in every quarter from 2018Q1 to 2022Q4 (first purchase by 2018Q1, last purchase in 2022Q4 or later), and adopters whose first wearable falls in 2019Q1–2021Q4, so the event window -4 to $+4$ is fully observed for every cohort. This leaves 457 adopters (358 clean) and 2,392 never-treated users. The comparison column re-estimates the unbalanced

panel on the same adopter cohorts (549 adopters, 4,086 controls). Reference period $t = -1$; post ATT averages quarters 1 to 4. Script: `design_did_balanced.R`. The conditional shoe-price outcome is dropped because it is not defined in every quarter.

Outcome	Balanced	Unbalanced (same cohorts)
Running shoe	-0.025 (0.012)	-0.017 (0.010)
Running-related (keyword)	-0.035 (0.019)	-0.031 (0.017)
Fitness gear	-0.001 (0.015)	-0.009 (0.014)
Supplements	0.027 (0.019)	0.018 (0.017)
Medication	-0.017 (0.016)	-0.011 (0.015)
PLACEBO: pet	-0.005 (0.017)	-0.013 (0.015)
PLACEBO: books	0.028 (0.021)	0.007 (0.019)
PLACEBO: gift cards	0.008 (0.016)	0.013 (0.014)
PLACEBO: cables/batteries	0.005 (0.023)	0.010 (0.021)
PLACEBO: cleaning	0.017 (0.017)	0.021 (0.015)
PLACEBO: grocery/food	0.018 (0.017)	0.018 (0.016)
General: $\ln(1+\text{\#purchases})$	0.001 (0.033)	-0.017 (0.032)
General: $\ln(1+\text{\$spend})$	-0.026 (0.053)	-0.047 (0.050)

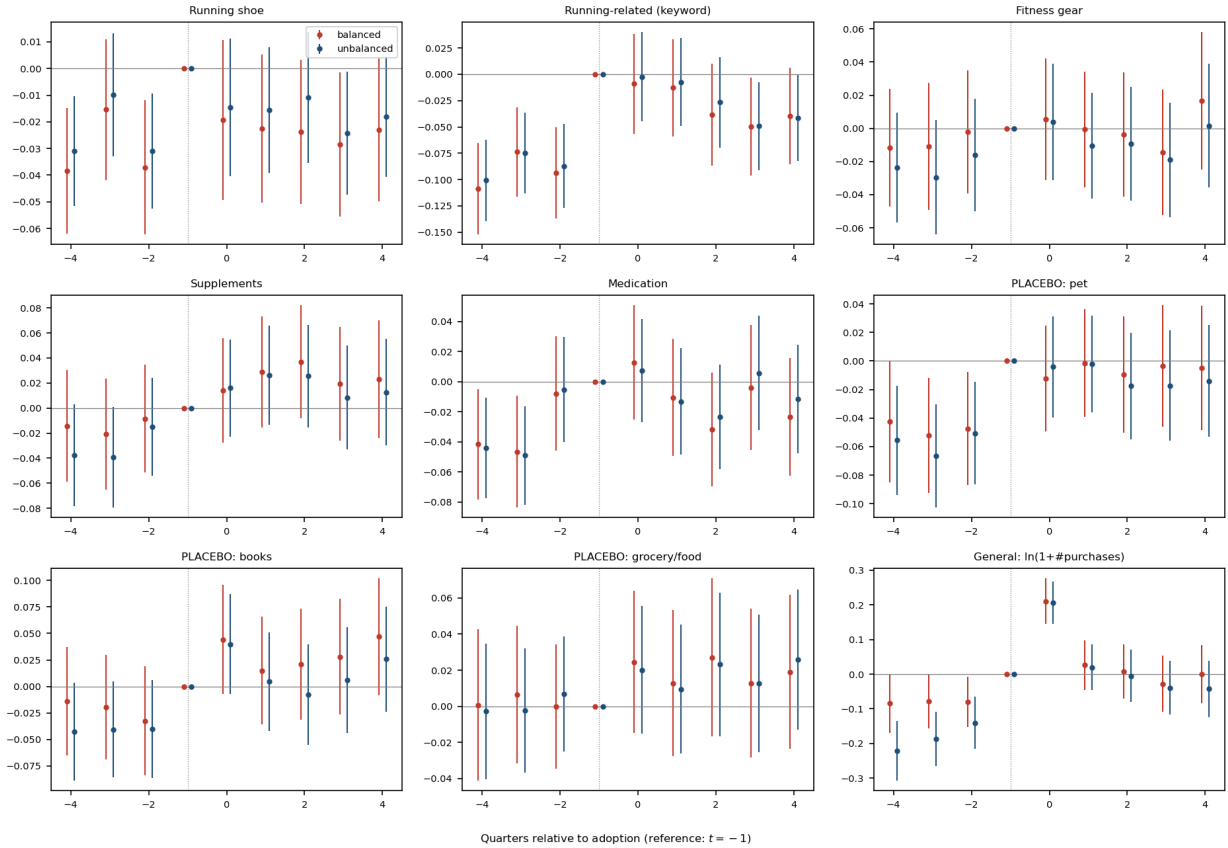


Figure 5: Event studies, balanced (red) vs. unbalanced on the same cohorts (blue).

- **The pre-trend does not go away.** Running-related purchases are about 0.09–0.11 below controls at $t = -4$ to -2 in the balanced panel, almost identical to the unbalanced one, and the pet placebo shows the same -0.05 pre-gap. Post ATTs barely move (running-related

−0.035 (0.019) vs. −0.031 (0.017); supplements +0.027 (0.019)). The clean sample behaves the same way (`design_tab_balanced_main_clean.tex`).

- **What this rules out.** Hypothesis 1 in its attrition form (adopters must remain active while controls can disappear) is not the main driver, since every user in the balanced panel is observed throughout. Total purchases $\ln(1+n)$ has a much smaller pre-trend in the balanced panel (≈ -0.08 vs. -0.14 to -0.22), so part of that run-up was composition, but the category-level pre-gaps remain. Hypotheses on latent intent, life events and income/tech affinity remain open.
- **Caveats.** The balanced sample selects long-lived heavy shoppers (about 3,100 of 5,000 users) and a short ± 4 quarter window; the post estimates are noisy (SEs of 0.01–0.02).

8 Conditioning on baseline lifestyle (doubly-robust DiD)

Approach. Callaway–Sant’Anna with `est_method="dr"` and baseline covariates, compared with the unconditional estimator on the *same* samples (reference $t = -1$, never-treated controls). Lifestyle proxies come from each user’s first 4 observed quarters (same rule for adopters and controls): log purchases, and 4 principal components (explaining 45% of variance) of log purchases, log spend, category shares (running, fitness, supplements, medication, pet, books, gift cards, cables, cleaning, grocery) and any-running/any-fitness indicators; plus survey age, income, household size, sex, college, white, diabetes, smoking, drinking and a survey-missing indicator (missing values mean-imputed; all 5,026 users matched). Scripts: `prep_design_covariates.py`, `design_did_lifestyle.R`.

- **Balanced panel:** as in the previous section (457 adopters, 2,392 controls; $e \in [-4, 4]$; post = quarters 1–4).
- **Unbalanced panel:** adopters need ≥ 6 quarters of pre-history so the baseline window precedes adoption (476 adopters, 4,086 controls; $e \in [-8, 8]$; post = quarters 1–8). This drops about 94 adopters relative to the earlier unbalanced runs.
- **Overlap is adequate:** the estimated adoption propensity averages 0.20 for adopters and 0.15 for balanced-panel controls (0.16 vs 0.10 unbalanced); the 99th percentile of control propensities is 0.39 and no adopter exceeds 0.9.
- **“Pre gap”** is the mean of the estimated pre-period coefficients ($e < -1$, relative to $t = -1$); it is the diagnostic that matters. A control for lifestyle works if placebo pre gaps move toward zero.

Balanced panel, all adopters

Outcome	Pre gap, uncond.	Pre gap, DR	Post ATT, uncond.	Post ATT, DR
Running shoe	-0.030	-0.032	-0.025 (0.012)	-0.025 (0.012)
Running-related (keyword)	-0.092	-0.093	-0.035 (0.019)	-0.035 (0.019)
Fitness gear	-0.008	-0.011	-0.001 (0.015)	0.000 (0.015)
Supplements	-0.014	-0.012	0.027 (0.019)	0.028 (0.019)
Medication	-0.032	-0.033	-0.017 (0.016)	-0.025 (0.016)
PLACEBO: pet	-0.047	-0.049	-0.005 (0.017)	-0.002 (0.017)
PLACEBO: books	-0.022	-0.026	0.028 (0.021)	0.034 (0.021)
PLACEBO: gift cards	0.013	0.011	0.008 (0.016)	0.013 (0.016)
PLACEBO: cables/batteries	-0.005	-0.004	0.005 (0.023)	0.008 (0.023)
PLACEBO: cleaning	0.003	0.006	0.017 (0.017)	0.017 (0.017)
PLACEBO: grocery/food	0.002	0.006	0.018 (0.017)	0.020 (0.017)
General: $\ln(1+\text{\#purchases})$	-0.081	-0.113	0.001 (0.033)	0.041 (0.034)
General: $\ln(1+\text{spend})$	-0.115	-0.173	-0.026 (0.053)	0.039 (0.052)

Unbalanced panel (adopters with ≥ 6 quarters pre), all adopters

Outcome	Pre gap, uncond.	Pre gap, DR	Post ATT, uncond.	Post ATT, DR
Running shoe	-0.027	-0.034	-0.019 (0.011)	-0.021 (0.017)
Running-related (keyword)	-0.082	-0.074	-0.040 (0.018)	-0.029 (0.029)
Fitness gear	-0.033	-0.037	-0.016 (0.015)	-0.011 (0.021)
Supplements	-0.048	-0.028	0.023 (0.019)	0.029 (0.026)
Medication	-0.043	-0.059	-0.010 (0.016)	-0.020 (0.025)
PLACEBO: pet	-0.060	-0.099	-0.028 (0.018)	-0.035 (0.028)
PLACEBO: books	-0.061	-0.086	-0.014 (0.021)	-0.002 (0.032)
PLACEBO: gift cards	0.001	0.012	-0.008 (0.016)	0.009 (0.022)
PLACEBO: cables/batteries	-0.051	-0.081	0.001 (0.022)	-0.032 (0.034)
PLACEBO: cleaning	-0.005	-0.020	0.029 (0.015)	0.011 (0.024)
PLACEBO: grocery/food	-0.009	0.012	0.016 (0.017)	0.035 (0.027)
General: $\ln(1+\text{\#purchases})$	-0.262	-0.300	-0.146 (0.038)	-0.055 (0.054)
General: $\ln(1+\text{spend})$	-0.377	-0.435	-0.276 (0.058)	-0.143 (0.082)

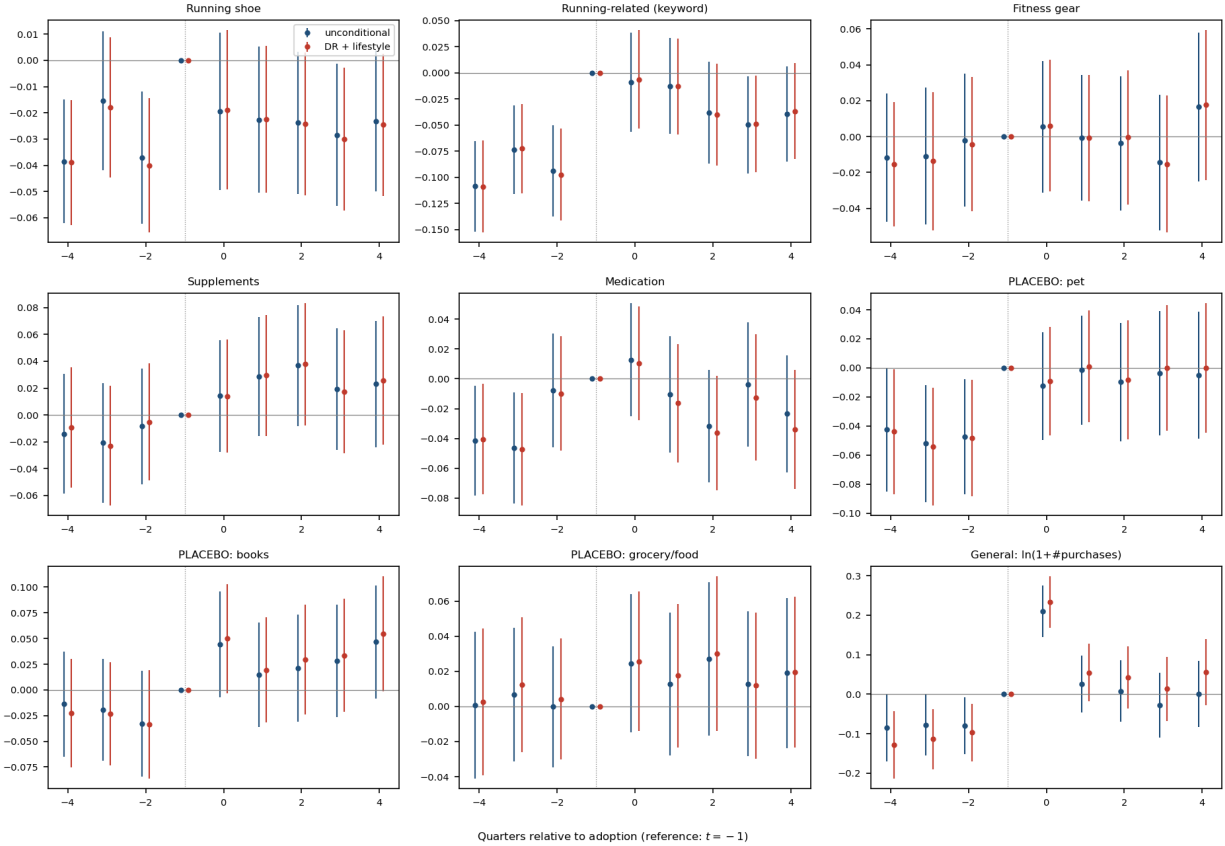


Figure 6: Balanced panel: unconditional (blue) vs. DR with lifestyle covariates (red).

Findings.

- **Baseline lifestyle does not remove the pre-trend.** In the balanced panel the running-related pre gap is -0.092 unconditional vs. -0.093 with covariates, the pet placebo -0.047 vs. -0.049 , books -0.022 vs. -0.026 ; the post ATTs are essentially unchanged (running-related -0.035 (0.019), supplements $+0.028$ (0.019), running shoe -0.025 (0.012)). In the

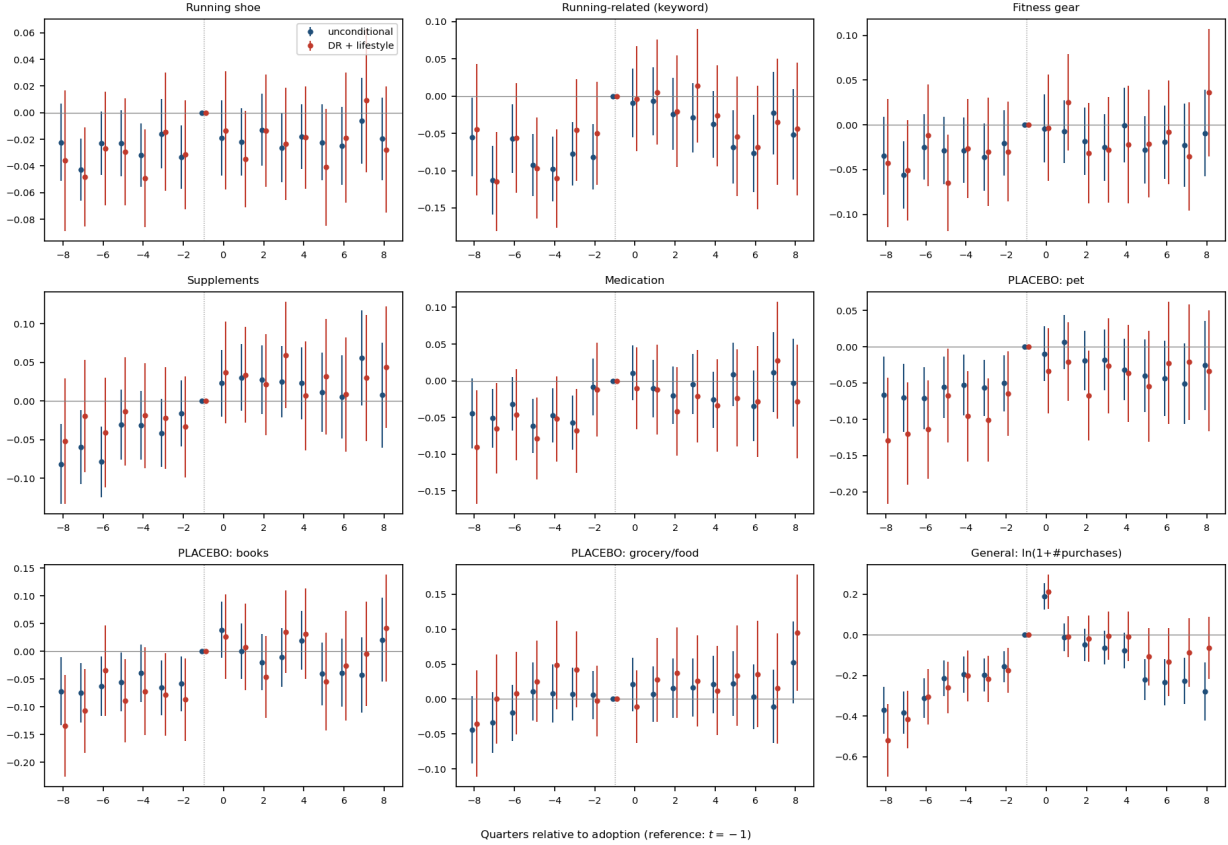


Figure 7: Unbalanced panel: unconditional vs. DR with lifestyle covariates.

unbalanced panel the gaps are similar or slightly larger in absolute value (running-related $-0.082 \rightarrow -0.074$, pet $-0.060 \rightarrow -0.099$), with larger standard errors.

- **By the criterion stated above, the control fails:** placebo pre-trends do not flatten, so the post estimates should not be read as causal effects of the device. The covariates do shift the general-intensity outcomes (balanced $\ln(1+n)$ post $0.001 \rightarrow 0.041$, unbalanced $-0.146 \rightarrow -0.055$), but those also have a steeper pre gap, so this mostly moves the level of the comparison, not the trend.
- **Interpretation.** Whatever drives the ramp is not captured by baseline (first-year) purchase mix or survey demographics. It is more likely a time-varying factor that evolves just before adoption (life events, an emerging fitness routine, hypotheses 3–4), so a time-invariant baseline adjustment cannot absorb it.
- **Clean sample:** same pattern (tables `design_tab_lifestyle_main_clean_*.tex`).

9 Pitfalls and follow-ups

- Pre-trends (and placebo pre-trends) mean the plain unconditional DiD is not credible; follow-ups: covariates/doubly-robust, not-yet-treated controls, HonestDiD bounds, matching on pre-adoption purchasing intensity (not run here).
- Quarterly indicators with sparse purchasers and a 10% (2018) vs 1% (2022) missing Category/Title rate make early outcomes undercounted; year effects not added.

- Bundlers are only 30 users, so the bundling comparison is underpowered; the ± 1 month window is a default.
- The conditional shoe-price placebo is estimated only on quarters with a shoe purchase (selected sample).